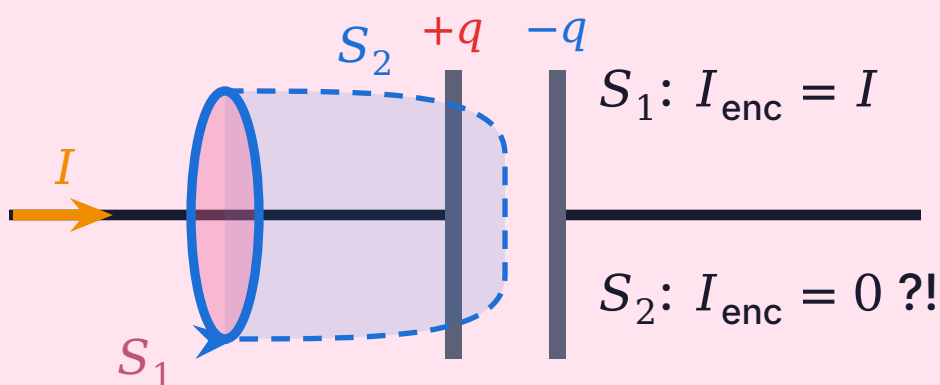


ELECTROMAGNETIC WAVES

Displacement Current • Maxwell

1 DISPLACEMENT CURRENT

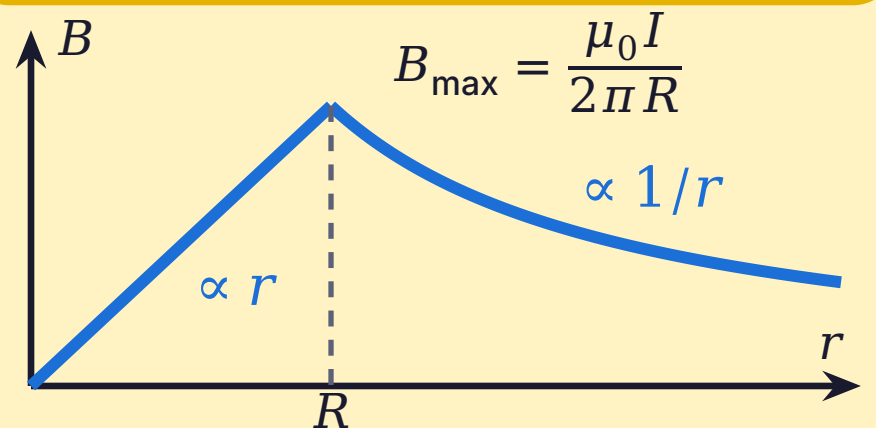
$$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$



Ampère fails in the gap → Maxwell adds I_d

2 B BETWEEN THE PLATES

$$B = \frac{\mu_0 I r}{2\pi R^2} \quad (r < R)$$

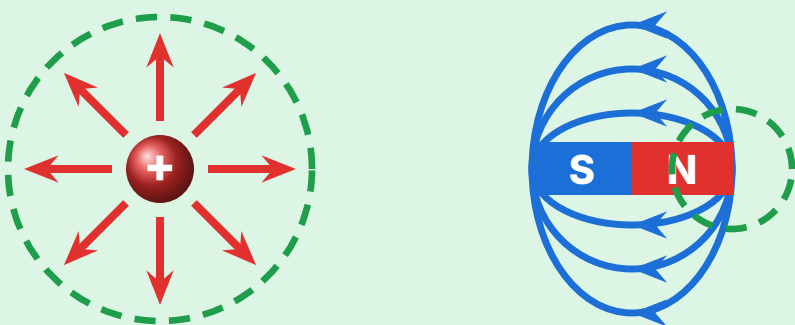


$I_d = I_c$ while charging; at steady DC $I_d = 0$

3 GAUSS'S LAWS (E & B)

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

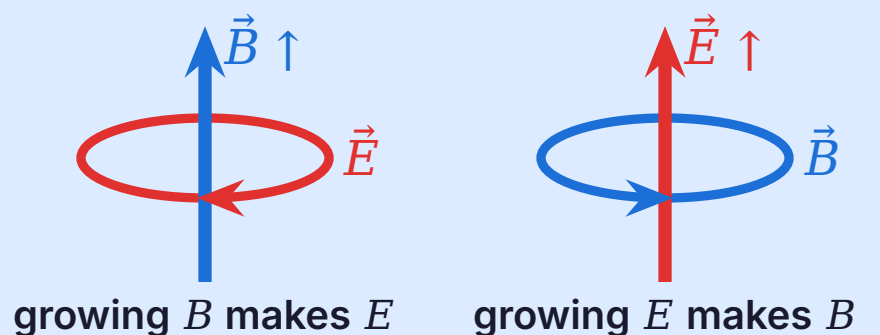


No monopoles: net flux of $\vec{B}$ is zero

4 FARADAY + AMPÈRE-MAXWELL

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + I_d)$$



growing B makes E

growing E makes B

Each changing field makes the other: a wave!

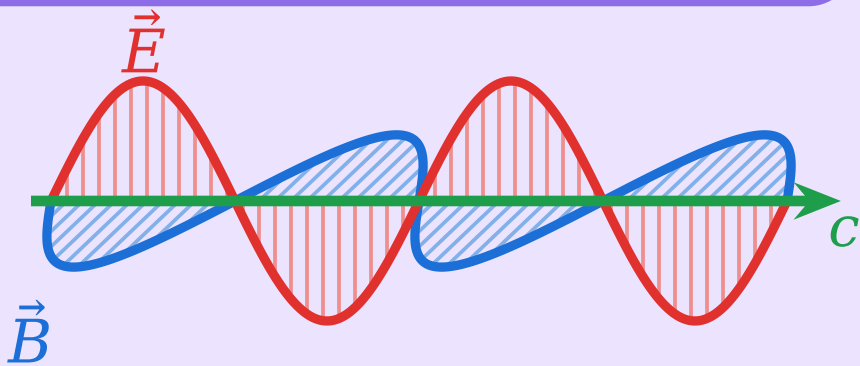
ELECTROMAGNETIC WAVES

Nature of EM Waves

5 THE EM WAVE IN 3D

$$E_y = E_0 \sin(kx - \omega t)$$

$$B_z = B_0 \sin(kx - \omega t)$$



$\vec{E} \perp \vec{B} \perp \vec{v}$ • in phase • transverse

6 AMPLITUDE RELATION

$$\frac{E_0}{B_0} = \frac{E}{B} = c$$

EXAMPLE

$$E_0 = 300 \text{ V m}^{-1}$$

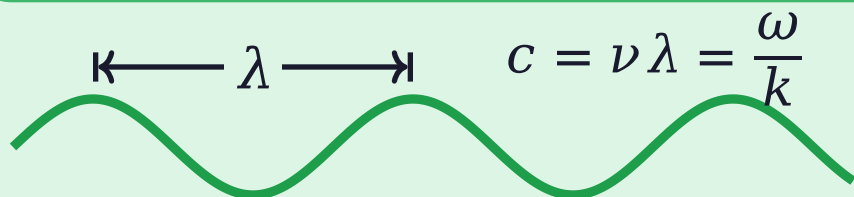
$$B_0 = \frac{E_0}{c} = \frac{300}{3 \times 10^8}$$

$$B_0 = 10^{-6} \text{ T} = 1 \mu\text{T}$$

At every instant $E = cB$ • B is tiny in SI

7 SPEED IN VACUUM

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$



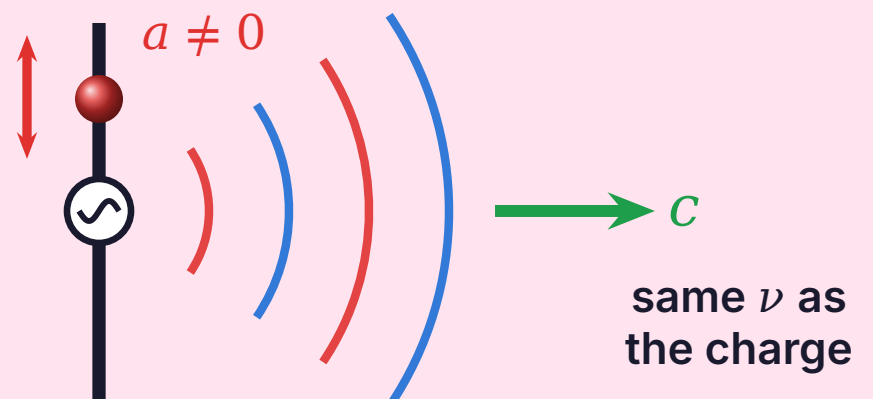
$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

Same c for ALL wavelengths in vacuum

8 SOURCE OF EM WAVES

$$\nu_{\text{wave}} = \nu_{\text{oscillating charge}}$$



Uniform velocity ($a = 0$) → NO radiation

ELECTROMAGNETIC WAVES

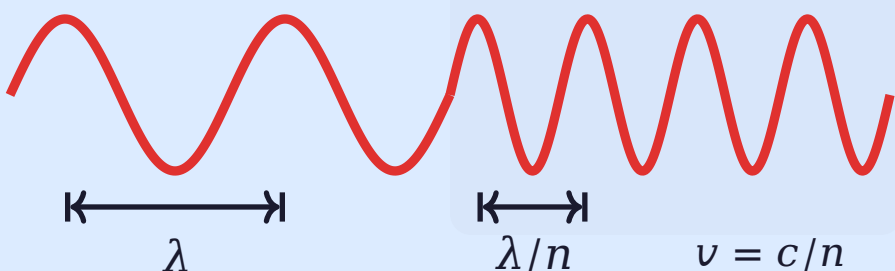
Medium • Refractive Index • Wave Equation

9

SPEED IN A MEDIUM

$$v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

vacuum

medium ($n = 2$)

v unchanged; v and λ both drop by n

10

REFRACTIVE INDEX

$$n = \frac{c}{v} = \sqrt{\mu_r \epsilon_r}$$

$$\mu_r = 1, \epsilon_r = 4$$

$$n = \sqrt{1 \times 4} = 2$$

$$v = \frac{c}{n} = 1.5 \times 10^8 \text{ m/s}$$

Non-magnetic medium ($\mu_r \approx 1$): $n \approx \sqrt{\epsilon_r}$

11

NUMERICAL: READ THE WAVE

$$B_y = B_0 \sin(kx + \omega t)$$

$$B_0 = 2 \times 10^{-7} \text{ T}$$

$$k = 500 \text{ m}^{-1}, \omega = 1.5 \times 10^{11} \text{ s}^{-1}$$

- $\lambda = 2\pi/k = 1.26 \text{ cm}$
- $\nu = \omega/2\pi \approx 2.4 \times 10^{10} \text{ Hz}$
- $E_0 = cB_0 = 60 \text{ V/m}$
- $+\omega t \Rightarrow$ moves along $-x$

$$E_z = 60 \sin(kx + \omega t) \text{ V/m} \quad (\hat{z} \times \hat{y} = -\hat{x})$$

12

KEY PROPERTIES

$$c = \nu \lambda$$

- ✓ need no medium (travel in vacuum)
- ✓ transverse $\rightarrow$ can be polarised
- ✓ carry energy and momentum
- ✗ not deflected by $\vec{E}$ or $\vec{B}$ fields

$$\text{FM 100 MHz: } \lambda = c/\nu = 3 \text{ m}$$

ELECTROMAGNETIC WAVES

Energy Density • Intensity

13 ENERGY DENSITY

$$u_E = \frac{1}{2} \epsilon_0 E^2$$

$$u_B = \frac{B^2}{2\mu_0}$$

$$B = \frac{E}{c} \Rightarrow \frac{B^2}{2\mu_0} = \frac{E^2}{2\mu_0 c^2} = \frac{1}{2} \epsilon_0 E^2$$

u_E 50 %

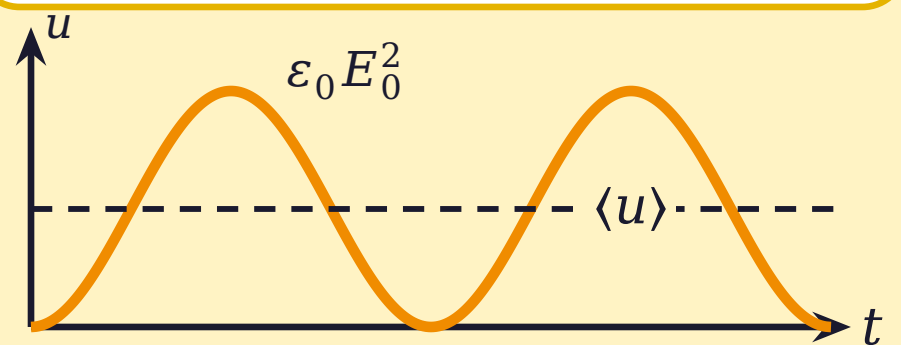
u_B 50 %

Energy is shared EQUALLY by E and B

14 TOTAL & AVERAGE ENERGY

$$u = u_E + u_B = \epsilon_0 E^2$$

$$\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2$$

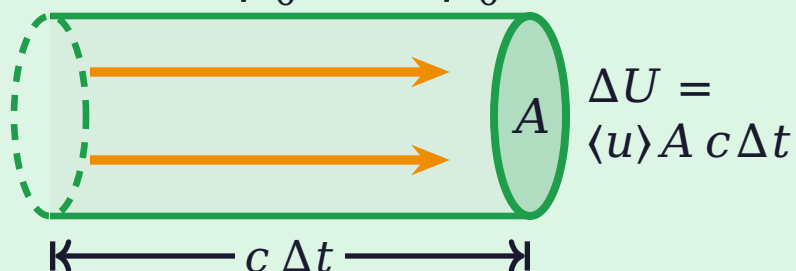


$\langle \sin^2 \rangle = \frac{1}{2} \rightarrow$ average is half the peak

15 INTENSITY

$$I = \langle u \rangle c = \frac{1}{2} \epsilon_0 c E_0^2$$

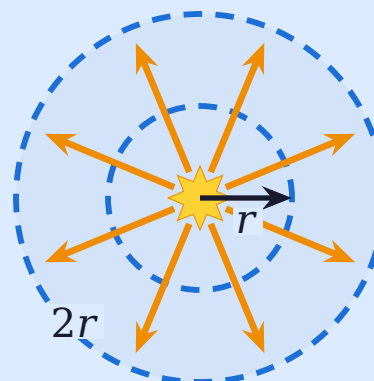
$$I = \frac{c B_0^2}{2\mu_0} = \frac{E_0 B_0}{2\mu_0}$$



I = energy per unit area per second (W m^{-2})

16 POINT SOURCE

$$I = \frac{P}{4\pi r^2}$$



$$r \rightarrow 2r$$

$$I \rightarrow I/4$$

$$E_0 \rightarrow E_0/2$$

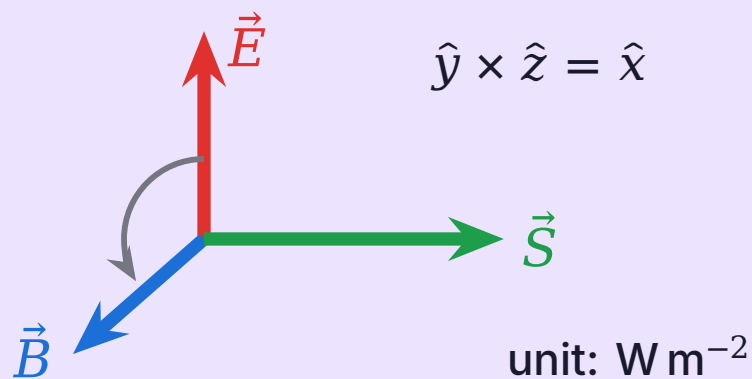
$I \propto 1/r^2$ but $E_0 \propto 1/r$ (as $I \propto E_0^2$)

ELECTROMAGNETIC WAVES

Poynting Vector • Momentum • Pressure

17 POYNTING VECTOR

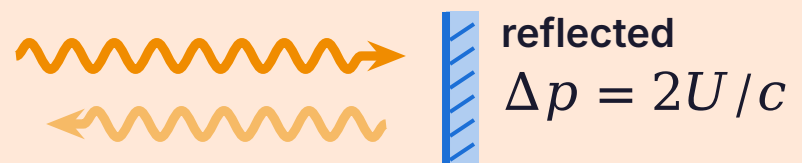
$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$



$\vec{S}$ points along travel • $|\langle \vec{S} \rangle| = I$

18 MOMENTUM OF EM WAVE

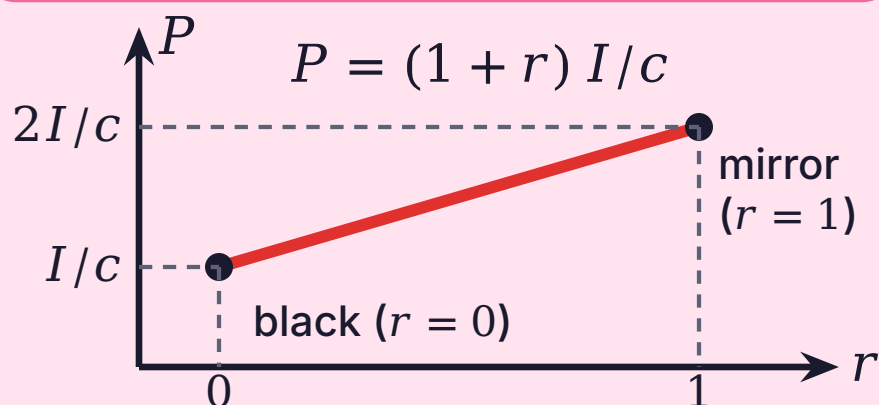
$$p = \frac{U}{c}$$



Force: $F = IA/c$ (absorb) • $2IA/c$ (reflect)

19 RADIATION PRESSURE

$$P = \frac{I}{c} \text{ or } \frac{2I}{c}$$

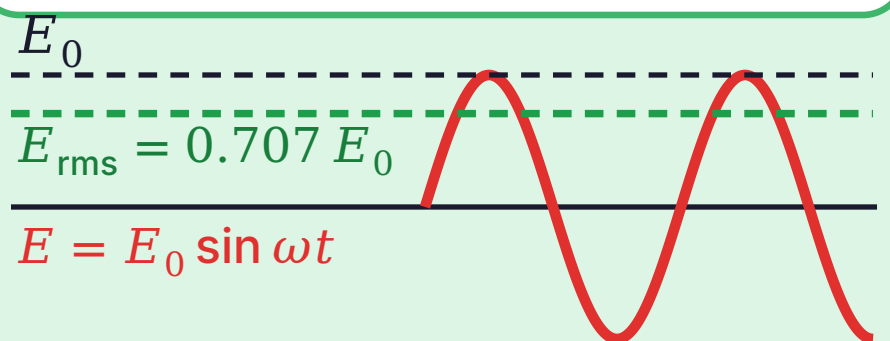


Sun, 1.4 kW/m^2 : $P = I/c \approx 4.7 \mu\text{Pa}$

20 RMS VALUES

$$E_{\text{rms}} = \frac{E_0}{\sqrt{2}}$$

$$I = \epsilon_0 c E_{\text{rms}}^2$$



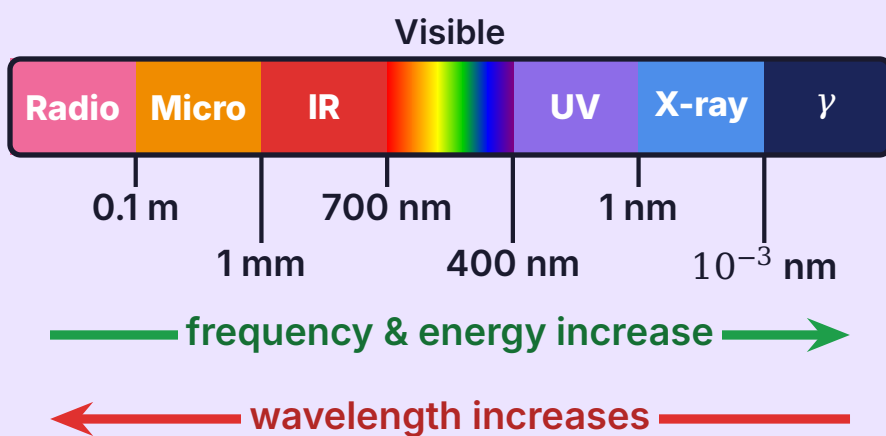
$E_0 = 100 \text{ V/m} \Rightarrow I \approx 13.3 \text{ W/m}^2$

ELECTROMAGNETIC WAVES

EM Spectrum • Order & Production

21 THE EM SPECTRUM

$$\lambda \downarrow \Rightarrow \nu \uparrow, E_{\text{photon}} \uparrow$$



λ : Radio > Micro > IR > Visible > UV > X > γ

22 RADIO WAVES

$$\lambda > 0.1 \text{ m}$$

Radio Micro IR Visible UV X-ray γ

SOURCE accelerated charges in conducting wires (antennas)

DETECTOR receiver's aerial

USES radio, TV; mobile phones (UHF band)

AM 530–1710 kHz • FM 88–108 MHz

23 MICROWAVES

$$1 \text{ mm} < \lambda < 0.1 \text{ m}$$

Radio Micro IR Visible UV X-ray γ

SOURCE klystron, magnetron, Gunn diode

DETECTOR point-contact diodes

USES radar (aircraft navigation), microwave oven

Oven: ν matched to water molecules

24 INFRARED (HEAT WAVES)

$$700 \text{ nm} < \lambda < 1 \text{ mm}$$

Radio Micro IR Visible UV X-ray γ

SOURCE hot bodies; vibrating atoms & molecules

DETECTOR thermopile, bolometer, IR photographic film

USES TV remote (IR LED), night vision, physiotherapy

Greenhouse: Earth's IR trapped by CO_2 , H_2O

ELECTROMAGNETIC WAVES

EM Spectrum • Detection & Uses

25 VISIBLE LIGHT

$$400 \text{ nm} < \lambda < 700 \text{ nm}$$

Radio Micro IR **Visible** UV X-ray γ

SOURCE electrons in atoms jumping to a lower energy level

DETECTOR the eye, photocells, photographic film

USES vision; photosynthesis in plants

$$\nu \approx 4 \times 10^{14} \text{ to } 7 \times 10^{14} \text{ Hz}$$

26 ULTRAVIOLET (UV)

$$1 \text{ nm} < \lambda < 400 \text{ nm}$$

Radio Micro IR Visible **UV** X-ray γ

SOURCE the Sun, very hot bodies, special lamps

DETECTOR photocells, photographic film

USES water purifiers (kill germs), LASIK eye surgery

Ozone layer absorbs most of the Sun's UV

27 X-RAYS

$$10^{-3} \text{ nm} < \lambda < 1 \text{ nm}$$

Radio Micro IR Visible UV **X-ray** γ

SOURCE X-ray tube: fast electrons hit a metal target

DETECTOR photographic film, Geiger tube, ionisation chamber

USES medical diagnosis (bones), cancer treatment

Over-exposure damages living tissue

28 GAMMA (γ) RAYS

$$\lambda < 10^{-3} \text{ nm}$$

Radio Micro IR Visible UV X-ray **γ**

SOURCE radioactive decay of nuclei; nuclear reactions

DETECTOR same as X-rays (Geiger tube, ionisation chamber)

USES medicine: destroy cancer cells

Highest frequency, most energetic photons

ELECTROMAGNETIC WAVES

Quick Revision • Save This!



FORMULA SHEET (1/2)

$$I_d = \varepsilon_0 d\Phi_E/dt$$

$$c = 1/\sqrt{\mu_0 \varepsilon_0} = 3 \times 10^8 \text{ m/s}$$

$$\nu = c/n, \quad n = \sqrt{\mu_r \varepsilon_r}$$

$$E_0 = c B_0, \quad c = \nu \lambda$$

$$\vec{v} \parallel \vec{E} \times \vec{B}$$

$$\vec{E} \perp \vec{B}, \text{ in phase}$$



FORMULA SHEET (2/2)

$$u_E = u_B, \quad \langle u \rangle = \frac{1}{2} \varepsilon_0 E_0^2$$

$$I = \frac{1}{2} \varepsilon_0 c E_0^2 = \varepsilon_0 c E_{\text{rms}}^2$$

$$I = P/4\pi r^2$$

$$\vec{S} = (\vec{E} \times \vec{B})/\mu_0$$

$$p = U/c \quad (\text{mirror: } 2U/c)$$

$$P = I/c \quad (\text{mirror: } 2I/c)$$



WHICH WAVE IS USED FOR?

Radar & microwave oven **Microwaves**

TV remote, night vision **Infrared**

FM radio, TV, mobiles **Radio waves**

Photosynthesis **Visible**

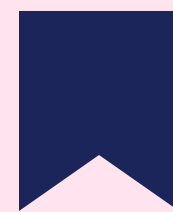
Water purifier, LASIK **Ultraviolet**

Bone fracture images **X-rays**

Killing cancer cells **Gamma rays**



DON'T LOSE THIS DECK



SAVE KARO

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